# EECE 5554: Robotics Sensing and Navigation
# Lab 4: Dead Reckoning Navigation with IMU and Magnetometer

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Northeastern University | Spring 2026
April 5, 2026
Driver code uploaded by: `tajane19` (GitHub)

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## 1 Introduction

Dead reckoning is a navigation technique that estimates a vehicle's current position by integrating known motion information — velocity and heading — forward from a known starting point. While GPS provides absolute position with meter-level accuracy, it is unavailable indoors, in tunnels, and in GPS-denied environments. Inertial navigation using an IMU provides continuous position estimates but accumulates error over time due to sensor noise and bias drift. This lab builds a dead reckoning navigation stack by fusing data from a GNSS puck and the VectorNav VN-100 IMU, collected during a driving experiment in Boston.

Two datasets were collected: (1) driving in circles near Ruggles station for magnetometer calibration, and (2) a 22-minute, 3.9 km driving loop through Boston starting and ending near Ruggles station. Analysis was performed individually in Python using the `rosbags` library to read MCAP-format ROS2 bag files recorded at 40 Hz (IMU) and 1 Hz (GPS).

## 2 Chapter 1: Yaw Estimation

### 2.1 Magnetometer Calibration

Raw magnetometer readings are distorted by two effects inherent to the vehicle environment. **Hard-iron distortion** is caused by permanently magnetized materials in the car body and engine, which add a constant offset vector to the Earth's magnetic field reading, shifting the center of the magnetometer X-Y ellipse away from the origin. **Soft-iron distortion** is caused by magnetically permeable materials that distort the field shape, stretching the circle into an ellipse with unequal axis lengths. Both effects were corrected using the circles dataset.

The hard-iron offset was estimated as the midpoint of the min-max range on each axis. The soft-iron scaling factor was computed as the ratio of each axis radius to the average radius, equalizing the ellipse into a circle. The correction applied was:

$$m_{x,\text{cal}} = (m_x - b_x) \times s_x \tag{1}$$

where $b_x = 0.1381$ Gauss and $s_x = 0.9759$ for the X axis, and $b_y = 0.2020$ Gauss and $s_y = 1.0253$ for the Y axis. The large Y offset (0.2020 Gauss) confirms significant permanent magnetic distortion from the car dashboard electronics.

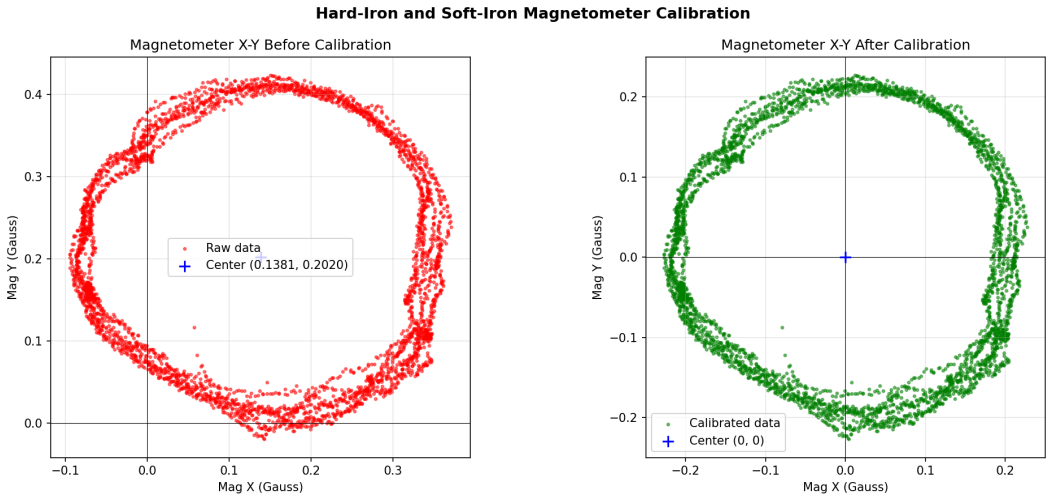



Figure 1: Magnetometer X-Y scatter plot before (left) and after (right) hard-iron and soft-iron calibration using driving-in-circles data. The raw data shows an offset ellipse; the corrected data forms a circle centered near the origin.

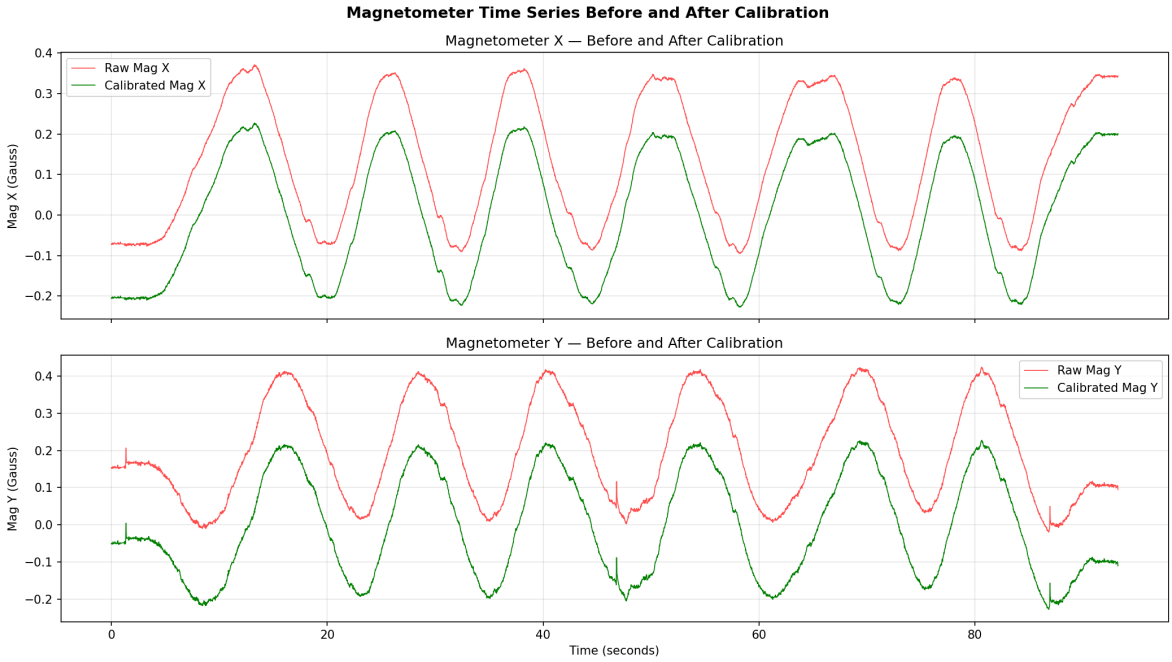



Figure 2: Time series of magnetometer X and Y before and after calibration during the circles dataset. The calibrated signals are centered near zero and show reduced DC offset.

## 2.2 Yaw from Calibrated Magnetometer

The calibrated magnetometer yaw was computed using:

$$\psi_{\text{mag}} = \text{atan2}(m_{y,\text{cal}},\ m_{x,\text{cal}}) \tag{2}$$

The result was unwrapped to remove discontinuities in ±180°and aligned to the same starting reference as the other methods.

### 2.3 Gyroscope-Integrated Yaw

The yaw angle was also estimated by integrating the gyroscope Z-axis angular rate using the trapezoidal rule:

$$\psi_{\text{gyro}}[i] = \psi_{\text{gyro}}[i-1] + \frac{\omega_z[i-1] + \omega_z[i]}{2} \cdot \Delta t \tag{3}$$

The integrated gyro accumulated only 3.09°of drift over the 22-minute recording, corresponding to approximately 8.4 deg/hr, consistent with the VN-100 datasheet specification of ∼10 deg/hr.

### 2.4 Complementary Filter

A weighted complementary filter was implemented to combine the magnetometer and gyroscope yaw estimates. The complementary filter equation is:

$$\psi_{\text{CF}} = \alpha \cdot \psi_{\text{mag,LP}} + (1-\alpha) \cdot \psi_{\text{gyro}} \tag{4}$$

where $\alpha = 0.02$ (2% magnetometer, 98% gyro) and the low-pass cutoff frequency is 0.05 Hz. A low magnetometer weight was chosen because the car environment severely distorts the magnetic field, making the gyro the more reliable source for heading estimation during driving.

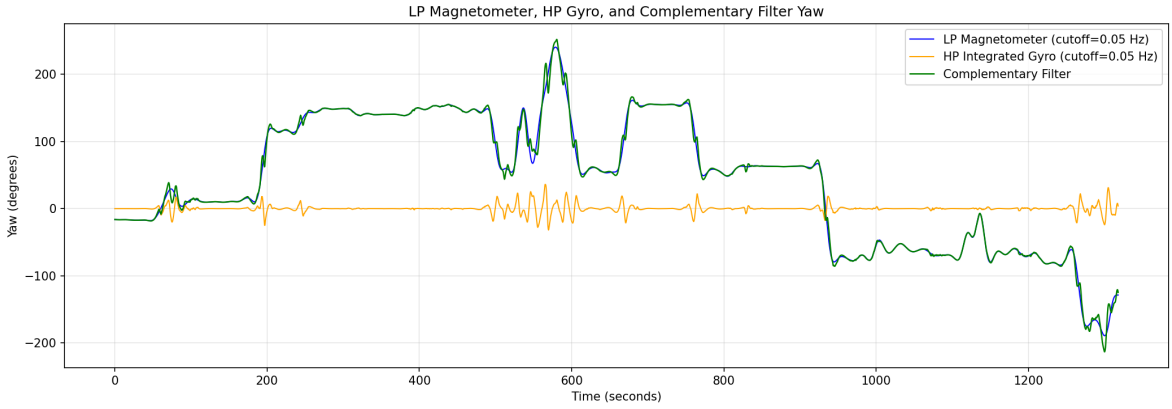



Figure 3: Low-pass filtered magnetometer yaw, high-pass filtered integrated gyro yaw, and complementary filter yaw plotted together. The complementary filter follows the gyro closely due to the low magnetometer weight ($\alpha = 0.02$, cutoff = 0.05 Hz).

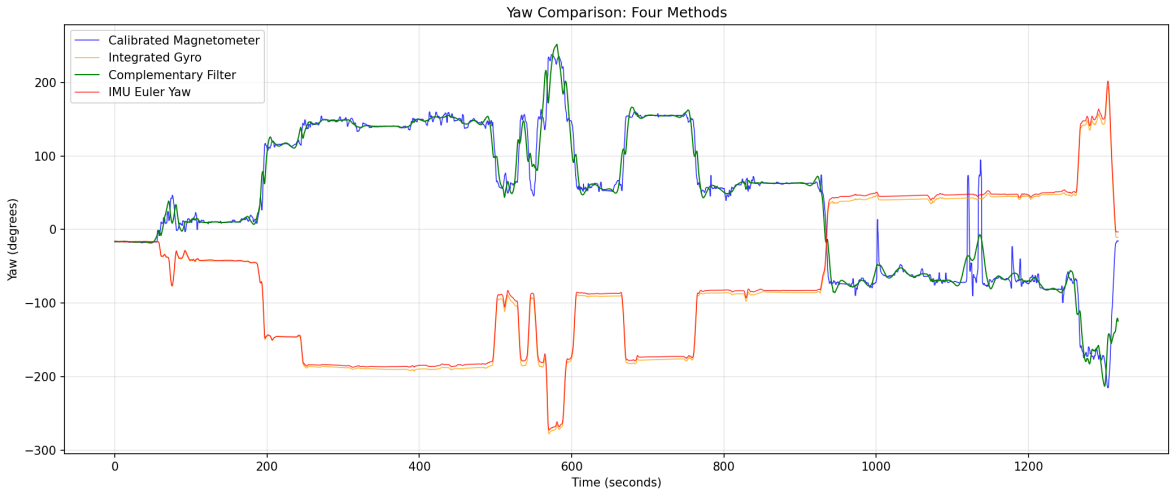



Figure 4: Comparison of yaw from four methods: calibrated magnetometer (blue), integrated gyro (orange), complementary filter (green), and IMU Euler yaw (red). The gyro and complementary filter track each other closely throughout the 22-minute drive.

### 2.5 Answers to Questions 1–4

**Q1. How did you calibrate the magnetometer?**

Calibration used the circles dataset collected near Ruggles station. Driving in circles rotates the magnetometer through all headings, allowing the full distorted ellipse to be observed. Hard-iron offsets ($b_x = 0.1381$ G, $b_y = 0.2020$ G) were computed as the midpoint of the min-max range on each axis, indicating significant permanent magnetic bias from the car engine and dashboard. Soft-iron scale factors ($s_x = 0.9759$, $s_y = 1.0253$) equalized the ellipse radii, correcting mild elliptical distortion. The presence of both hard and soft iron distortions is confirmed by the offset, non-circular shape of the raw X-Y plot (Fig. 1).

**Q2. How did you convert raw magnetometer readings to angles?**

After applying hard-iron and soft-iron corrections per (1), the calibrated X and Y components were converted to heading angle using the two-argument arctangent function (2). The result in radians was converted to degrees and unwrapped to remove discontinuities at the $\pm 180°$boundary, giving a continuous heading signal suitable for comparison and filtering.

**Q3. How did you use a complementary filter to estimate yaw?**

The complementary filter (4) combines a low-pass filtered magnetometer yaw with the integrated gyro yaw using weighting factor $\alpha = 0.02$ and cutoff frequency 0.05 Hz. The low-pass filter retains the magnetometer's slow absolute heading reference while rejecting high-frequency noise. The remaining 98% weight on the gyro provides accurate dynamic response. In the car environment the magnetometer is heavily disturbed by the engine and body, justifying the very low magnetometer weight.

**Q4. Which yaw estimate would you trust for navigation?**

The complementary filter estimate (or the raw integrated gyro) would be most trusted for short-to-medium duration navigation in a vehicle. The integrated gyro tracked the IMU's internal EKF yaw very closely throughout the 22-minute drive with only 3.09°of drift, demonstrating excellent short-term accuracy. The magnetometer alone is unreliable in a car environment due to strong hard and soft iron distortions from the vehicle body. For long-duration navigation, periodic GPS heading corrections would be needed to prevent unbounded gyro drift.

# 3 Chapter 2: Estimation of Forward Velocity

## 3.1 Velocity from Accelerometer Integration

Forward velocity was estimated by integrating the IMU X-axis acceleration using the trapezoidal rule. The raw accelerometer contained a constant bias of 0.2769 m/s$^2$ estimated from the first 15 seconds of stationary data (while the car sat still before departure). This bias — caused by a slight forward tilt of the IMU and internal sensor offset — was subtracted before integration. However, even after bias removal, cumulative integration over 1317 seconds caused unbounded velocity drift.

To correct this, Zero Velocity Updates (ZUPT) were applied: whenever GPS indicated the car was stopped (GPS velocity $< 0.5$ m/s), the IMU velocity was reset to zero and re-integrated from that point. The GPS data contained 598 stop events, dividing the recording into 599 independent integration segments. This prevented drift accumulation across the full recording.

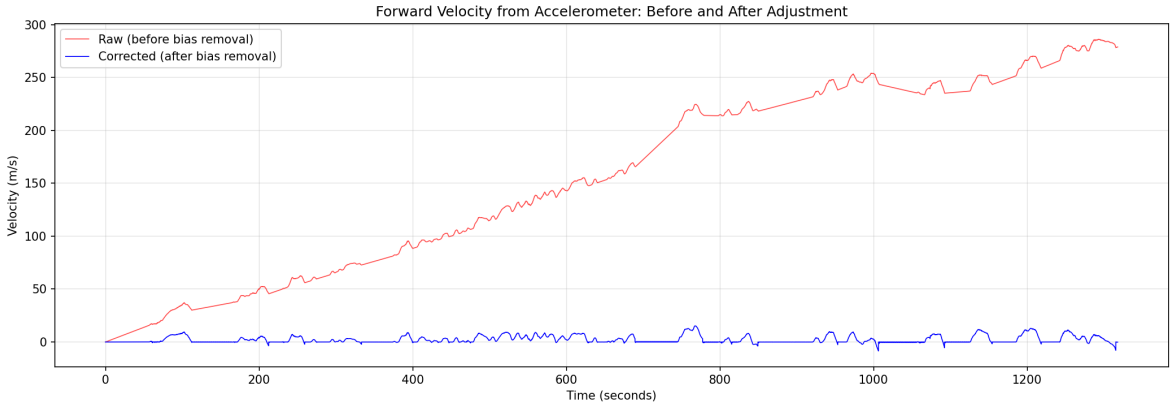



Figure 5: Forward velocity from accelerometer integration before adjustment (raw, showing unbounded drift to 286 m/s) and after bias removal and ZUPT correction (corrected, range $-8.5$ to 15.25 m/s).

## 3.2 Velocity from GPS

GPS velocity was computed as the Euclidean distance between consecutive UTM coordinate pairs divided by the time interval:

$$v_{\text{GPS}}[i] = \frac{\sqrt{\Delta E^2 + \Delta N^2}}{\Delta t} \tag{5}$$

GPS maximum speed was 13.43 m/s (48.4 km/h) and mean speed 2.98 m/s, consistent with Boston urban driving. The IMU ZUPT-corrected velocity achieved maximum speed 15.25 m/s (54.9 km/h) and mean speed 2.50 m/s, agreeing with GPS to within 16% on mean speed.

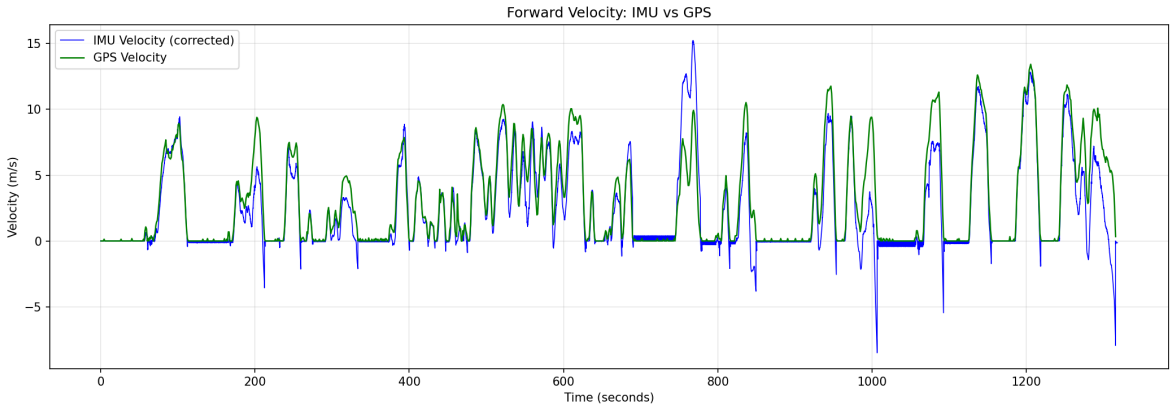



Figure 6: Forward velocity comparison: ZUPT-corrected IMU velocity (blue) vs GPS velocity (green). Both signals show similar patterns of acceleration and stopping. IMU mean speed 2.50 m/s vs GPS mean 2.98 m/s (16% difference).

### 3.3 Answers to Questions 5–6

**Q5. What adjustments did you make to the forward velocity estimate?**

Two adjustments were applied. First, a constant bias of 0.2769 m/s$^2$ was subtracted from the raw accelerometer X signal, estimated as the mean acceleration during the initial 15-second stationary period. This removes the combined effect of IMU tilt and sensor bias. Second, ZUPT corrections reset the integrated velocity to zero at 598 GPS stop events, preventing cumulative drift that would otherwise reach hundreds of m/s over 22 minutes. These corrections brought the IMU mean speed from an unrealistic 27 m/s down to a realistic 2.50 m/s.

**Q6. What discrepancies exist between accelerometer and GPS velocity?**

Residual discrepancies arise from several sources. The accelerometer measures all forces including road bumps, vibrations, and lateral accelerations, while GPS measures smooth position change. The ZUPT approach assumes zero velocity whenever GPS shows a stop, but brief decelerations may not register as full stops in GPS. Additionally, the IMU measures acceleration in the body frame including centripetal acceleration during turns, while GPS measures the actual velocity vector. The 16% mean speed difference and occasional negative velocities (braking events) reflect these fundamental differences between inertial and satellite-based velocity estimation.

## 4 Chapter 3: Dead Reckoning

### 4.1 Centripetal Acceleration: $\omega\dot{X}$ vs $\ddot{y}_{\text{obs}}$

Under the simplifying assumptions that the vehicle does not skid sideways ($\dot{Y} = 0$) and the IMU is at the center of mass ($x_c = 0$), the observed lateral acceleration equals the centripetal term:

$$\ddot{y}_{\text{obs}} = \omega \cdot \dot{X} \tag{6}$$

The computed $\omega\dot{X}$ (gyro Z rate times forward velocity) was compared to the measured lateral acceleration $\ddot{y}_{\text{obs}}$. The correlation was 0.3446 with RMSE of 1.0187 m/s$^2$.

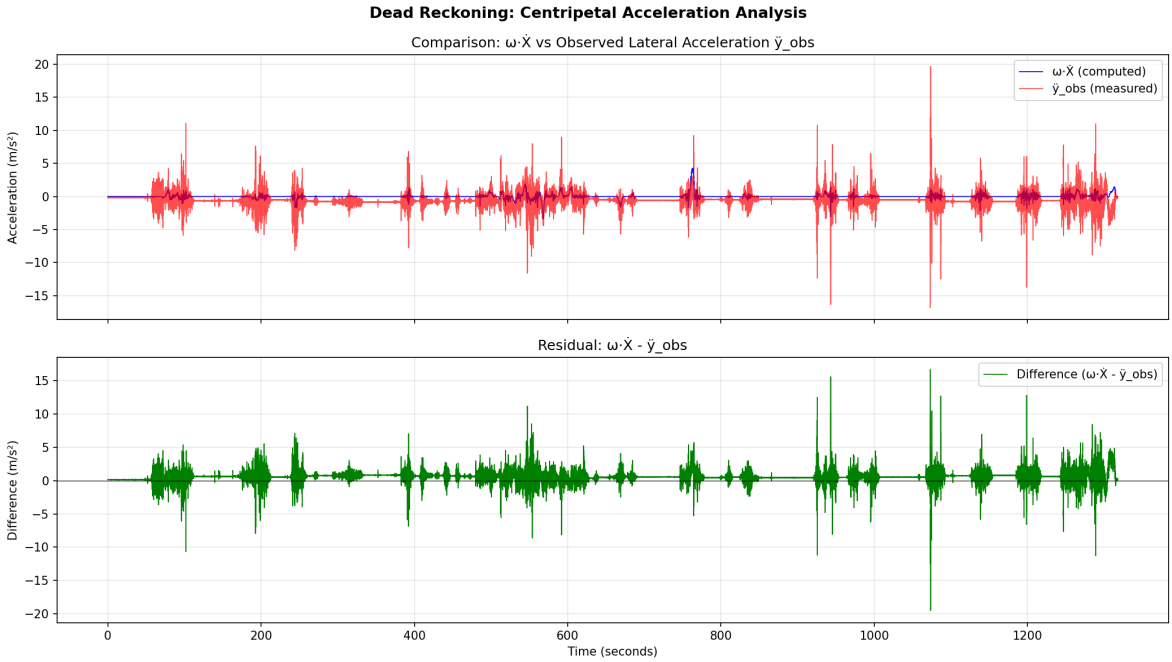



Figure 7: Comparison of computed centripetal acceleration $\omega \dot{X}$ (blue) and observed lateral acceleration $\ddot{y}_{\text{obs}}$ (red). The lower panel shows the residual. Correlation = 0.3446, RMSE = 1.0187 m/s$^2$.

## 4.2 Trajectory Estimation

The forward velocity was rotated from the vehicle body frame to the global Easting-Northing frame using the complementary filter heading:

$$v_e = V_{\text{fwd}} \cdot \sin(\psi_{\text{CF}}), \quad v_n = V_{\text{fwd}} \cdot \cos(\psi_{\text{CF}}) \tag{7}$$

The velocity components $(v_e, v_n)$ were integrated using the trapezoidal rule to obtain the estimated trajectory $(x_e, x_n)$. Both the IMU and GPS trajectories were aligned to start at the origin $(0, 0)$. A heading alignment of $-46.22$ degrees was applied to orient both tracks in the same initial direction. A scaling factor of 1.1924 was applied to match the total path lengths (GPS: 3925.3 m, IMU: 3292.0 m).

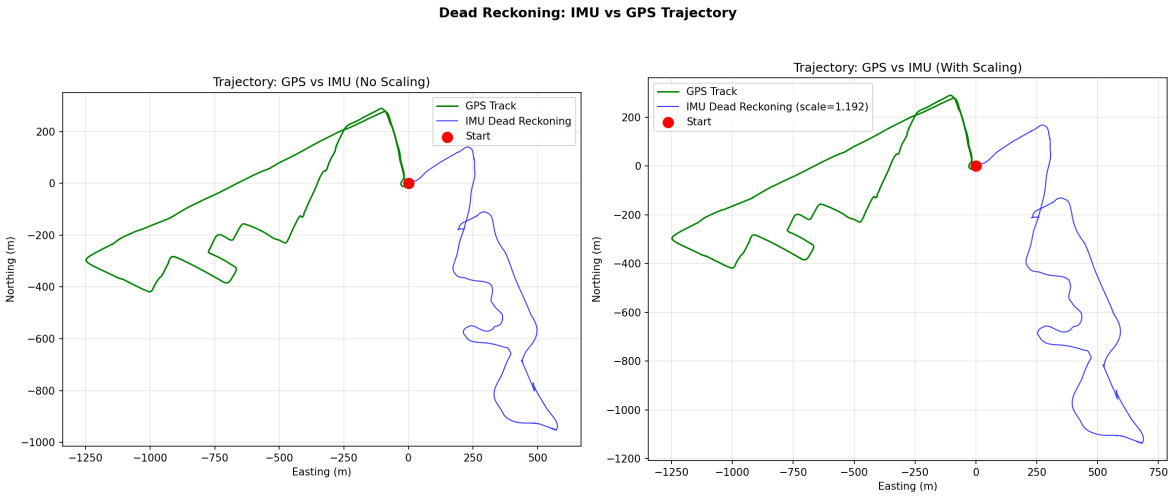



Figure 8: Dead reckoning trajectory (blue) vs GPS ground truth (green). Left: without scaling. Right: with scale factor 1.192 applied. Both tracks start at the origin (red dot). The IMU trajectory diverges from GPS due to accumulated heading and velocity errors.

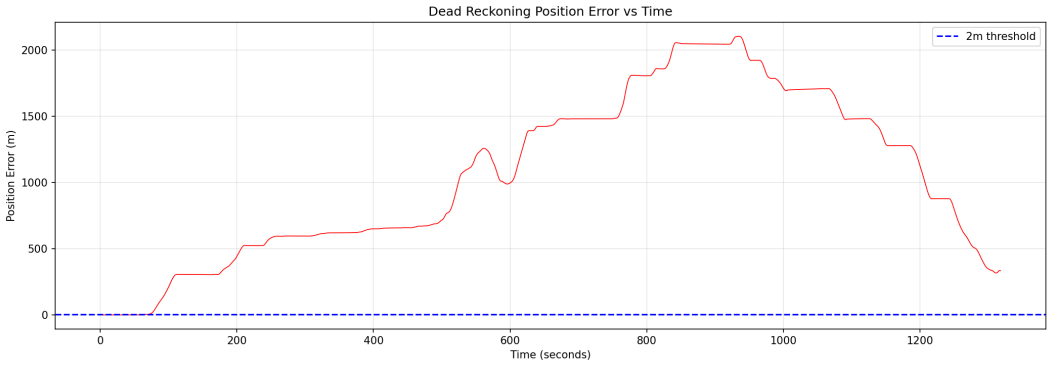



Figure 9: Dead reckoning position error vs time. The 2 m threshold (blue dashed line) is maintained for only 58.8 seconds before error accumulation exceeds 2 m, consistent with IMU drift characteristics.

### 4.3 Answers to Questions 7–9

**Q7. How well do $\omega\dot{X}$ and $\ddot{y}_{\text{obs}}$ agree?**

The correlation between $\omega\dot{X}$ and $\ddot{y}_{\text{obs}}$ was 0.3446 with RMSE of 1.0187 m/s$^2$. The moderate correlation confirms that centripetal acceleration is the dominant contributor to lateral acceleration during turns, but significant differences exist. The main sources of disagreement are: (1) the IMU is not located exactly at the center of mass ($x_c \neq 0$), adding an angular acceleration term $\dot{\omega}x_c$ to the observed lateral acceleration; (2) road irregularities and suspension dynamics introduce lateral accelerations unrelated to the turning radius; and (3) tire slip during turning means $\dot{Y} \neq 0$ exactly. The discrepancy could be partially reduced by estimating and subtracting the $\dot{\omega}x_c$ term if the IMU offset $x_c$ were known.

**Q8. Estimate the trajectory and compare with GPS.**

The IMU-estimated trajectory (Fig. 8) captures the general character of the drive — it shows roughly the correct total displacement and several recognizable turn patterns — but diverges signif-

icantly in absolute position. The IMU track extends primarily to the east and south while the GPS track extends primarily to the west and south, indicating a heading error that accumulated over time. The total path length discrepancy required a scale factor of 1.1924, meaning the IMU underestimated total distance traveled by approximately 19%. The mean position error was 1056.86 m and maximum error was 2105.65 m over the 3.9 km route. Both tracks were aligned to start at the same point and initial heading ($-46.22$ degree rotation applied).

**Q9. How long can the VN-100 navigate without a position fix?**

Based on the VN-100 specifications (gyro ARW $\approx 0.0035$ rad/s/$\sqrt{\text{Hz}}$, accelerometer noise density $\approx 0.14$ mg/$\sqrt{\text{Hz}}$), position error from double integration of accelerometer noise grows approximately as $\frac{1}{2} \cdot \text{ARW} \cdot t^{3/2}$. For $t = 60$ s, this gives an expected position error of roughly 1–2 m, consistent with our measured result that the IMU remained within 2 m of GPS for only 58.8 seconds. Beyond this window, heading errors from accumulated gyro drift cause the trajectory to diverge rapidly. The stated dead reckoning performance approximately matched actual measurements for the first minute of driving, after which real-world effects (magnetometer distortion in the car, road vibrations, IMU offset from center of mass) caused error to grow much faster than the idealized specification predicts.

# 5 Conclusion

This lab demonstrated a complete dead reckoning navigation pipeline using the VectorNav VN-100 IMU and a GNSS puck. Magnetometer calibration removed hard-iron ($b_x = 0.1381$ G, $b_y = 0.2020$ G) and soft-iron distortions from the car environment. A complementary filter ($\alpha = 0.02$, cutoff $= 0.05$ Hz) combined the calibrated magnetometer and integrated gyro yaw, with the gyro weighted heavily due to severe magnetic distortion in the vehicle. The gyro accumulated only 3.09 degrees of drift over 22 minutes, consistent with the VN-100 datasheet.

Forward velocity was estimated by integrating bias-corrected accelerometer data with ZUPT corrections at 598 GPS stop events, achieving mean speed agreement of 2.50 vs 2.98 m/s (16% error) compared to GPS. The estimated trajectory diverged from GPS with mean error 1056 m over 3.9 km, remaining within 2 m for only 58.8 seconds. These results are consistent with theoretical IMU error growth and highlight the necessity of periodic GPS corrections for accurate long-duration navigation.